

Abu Ubaida

Full-Stack & AI Engineer

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SUMMARY

Results-driven Full-Stack Software & AI Engineer with 3+ years of experience architecting scalable backend applications, microservices, and AI-powered products. Promoted twice in under three years (Junior to Senior Software Engineer). Expert in Python (FastAPI, Django) with a track record of optimizing system performance, cutting manual workflows, and shipping high-performance APIs and production AI agents.

EXPERIENCE

Senior Software Engineer

[Smartbenefits](#) · Karachi, Pakistan

Jul 2026 – Present

- Architected a nightly Celery-scheduled Lambda analytics pipeline running complex SQL against a read-only replica, pre-computing usage stats for 600+ enterprise clients — cutting dashboard load times from ~15 minutes to under 10ms with zero production DB load.
- Cut LLM token usage 80% (7,800 to 1,500 tokens/client) for AI-generated chart insights by restructuring raw query output into a compact delta array before inference, across 600+ nightly dashboard refreshes.
- Built an automated JSON export pipeline to Amazon S3 with a cached API (1-hour TTL) powering interactive charts in the HR Portal, supporting real-time renewal conversations for the Sales team.
- Mentored 2 junior engineers, instituting PR standards and structured code reviews that reduced review cycle time.

Software Engineer

[Smartbenefits](#) · Karachi, Pakistan

Jan 2025 – Jul 2026

- Built a multi-agent AI chatbot (Vertex AI, ADK, WebSocket) resolving 1,000+ monthly medical queries at 85%+ accuracy — with self-updating agents that auto-enrich the knowledge base on every OPD claim settlement.
- Led end-to-end architecture of Hemaayah, Pakistan's first remittance-linked health insurance product — built 3 platforms with WhatsApp-based policy issuance and webhook integrations (Vodafone Qatar, Ompay, Bank of Punjab Exchange), enabling fully automated, zero-touch policy delivery.
- Architected serverless microservices (AWS Lambda) to offload bulk exports, cron jobs, and heavy I/O — eliminating recurring memory spikes and cutting monthly compute spend.
- Reduced HR workload by 30% by building in-portal OPD claims submission and Celery-based automated onboarding/offboarding.
- Shipped a branch-level oversight dashboard and quarterly NPS system delivering actionable insights to 1,000+ HR professionals.

Junior Software Engineer

[Smartbenefits](#) · Karachi, Pakistan

Jan 2024 – Jan 2025

- Automated quarterly utilization reporting and policy-expiry notifications for 400+ corporate clients via Celery, eliminating 20+ hours/week of manual HR effort.
- Built the Provident Fund Management feature enabling real-time dividend and fund-growth tracking, reducing PF-related support queries.
- Delivered a doctor chat feature in partnership with NoorCare, driving 500+ chats/week.
- Built a post-claim-settlement feedback questionnaire tool generating 1,000+ weekly responses.

Founder / Independent Projects

SaaS & Open Source · Remote

2024 – Present

- Designed and built Feesday, a B2B SaaS platform automating recurring-fee collection — migrated the reminder engine to a serverless architecture (Celery ! SQS ! AWS Lambda ! API Gateway) handling 1M+ requests reliably.
- Architected AddaZakat, a service-oriented platform (Django, FastAPI, Next.js) connecting global donors to verified NGOs, with a LangGraph/MCP conversational AI assistant and 3,000+ monthly visitors.

FEATURED PROJECTS

- Feesday — B2B payment-reminder & reconciliation engine (FastAPI, Next.js, PostgreSQL, WhatsApp API). Serverless Celery ! SQS ! Lambda pipeline; 1M+ requests with ms-level responses.
- AddaZakat — AI-agent social-impact platform (Next.js, Django, FastAPI, LangGraph, FastMCP). LangGraph/MCP story-publishing agent; 3,000+ monthly visitors and 17,000+ views in 90 days.
- Hemaayah — Multi-portal policy generation & validation backend (Django, WhatsApp API, Botpress). Three portals with automated zero-touch policy issuance.
- Super School — Asynchronous GraphQL backend (FastAPI, Strawberry GraphQL, PostgreSQL, Docker). Queue-driven multi-tenant school administration.

AWARDS & RECOGNITION

- Breakout Performer Award — Smartbenefits (March 2026): honored for exceptional technical contributions and rapid deployment of multiple end-to-end AI and backend features.

EDUCATION

- B.S. in Software Engineering (EQF Level 6) — University of Karachi (UBIT), 2024. Focus: Distributed Systems, Algorithms, Software Design, Capstone.

CERTIFICATIONS

- LLM Engineering, RAG and AI Agents — Udemy (April 2026)
- FastAPI Full Stack Mastery — Udemy (November 2025)
- Problem Solving (Advanced) — HackerRank (July 2024)

TECHNICAL SKILLS

Backend & Systems: Python, FastAPI, Django, Flask, Node.js, GraphQL, REST, Celery, Redis, PostgreSQL, MySQL, MongoDB

AI & Agentic Workflows: LangGraph, Vertex AI, RAG Pipelines, FastMCP, Google GenAI SDK, Ollama, ADK, LangChain

Frontend Systems: TypeScript, React.js, Next.js, Tailwind CSS, HTMX, State Management

Cloud & DevOps: AWS (Lambda, EC2, S3, EventBridge, CloudFront), Docker, Nginx, GitHub Actions, CI/CD